

Multiple Choice

1. **Answer:** D

Options: A) 1.5 R; B) 2 R; C) 2.5 R; D) 3.5 R

Note: The correct answer is 3.5 R because at high temperatures, a diatomic gas has 5 degrees of freedom (3 translational and 2 rotational), leading to a molar heat capacity at constant volume of $C_V = \frac{5}{2}R$ for translational and rotational contributions, plus an additional $\frac{1}{2}R$ contribution from vibrational modes that become active, resulting in a total of $C_V = 3.5R$.

2. **Answer:** D

Options: A) Ammonium hydroxide; B) Sulfuric acid; C) Acetic acid; D) Potassium hydrogen phthalate

Note: Potassium hydrogen phthalate is a primary standard because it is a pure, stable compound that can be accurately weighed and used to determine the concentration of a base in titrations due to its known and consistent acidity.

3. **Answer:** A

Options: A) 3.74 T; B) 5.19 T; C) 6.08 T; D) 9.49 T

Note: The correct answer of 3.74 T corresponds to the magnetic field strength at which the rotational correlation time of 1 ns leads to optimal spin-lattice relaxation, as determined by the relationship between magnetic field and relaxation times in NMR spectroscopy.

4. **Answer:** D

Options: A) 10.8×10^{-5} ; B) 4.11×10^{-5} ; C) 3.43×10^{-5} ; D) 1.71×10^{-5}

Note: The correct answer of 1.71×10^{-5} is derived from the Boltzmann distribution, which relates the population of a state to its energy relative to the ground state.

4. **Answer:** A

Options: A) HCl; B) AgCl; C) CaCl₂; D) CCl₄

Note: HCl has the lowest melting point because it is a polar covalent molecule with relatively weak dipole-dipole interactions, compared to the stronger ionic or hydrogen bonding interactions present in other compounds.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because thallium's +1 oxidation state is more stable due to the inert pair effect, where the s-electrons are less likely to participate in bonding, making the +3 state less favorable.

7. **Answer:** True

Options: A) True; B) False

Note: The thermal stability of group-14 hydrides increases in that order because as you move from CH₄ to PbH₄, the bond strength of the carbon-hydrogen bonds decreases, while the heavier hydrides exhibit weaker bonds due to the increasing size and decreasing electronegativity of the central atom, leading to greater instability and easier decomposition.

8. **Answer:** False

Options: A) True; B) False

Note: Silicon predominantly occurs in nature as a silicate or silicon dioxide rather than as a metallic element, making the statement false.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because AlCl₃ dissociates in solution into one aluminum ion (Al³⁺) and three chloride ions (Cl⁻), resulting in a total of four ions per formula unit, which contributes to a higher ionic strength compared to other solutions with fewer ions.

10. **Answer:** False

Options: A) True; B) False

Note: Planck's quantum theory is primarily used to explain the emission spectra of atoms, particularly monatomic gases, rather than diatomic molecules, which require more complex quantum mechanical models due to their vibrational and rotational transitions.

Long Form Response

11. **Answer:** The ¹H NMR spectrum of CHCl₃ appears as a singlet due to the presence of chlorine isotopes, specifically ³⁵Cl and ³⁷Cl, both of which have electric quadrupole moments. In NMR spectroscopy, the presence of quadrupole nuclei can lead to broadening of the signals, but in the case of CHCl₃, the two chlorine isotopes are present in such a way that they do not lead to splitting of the proton signal. Instead, the magnetic environment around the hydrogen atom is effectively averaged out by the two isotopes, resulting in a single peak. Furthermore, the quadrupole interactions are not strong enough to cause noticeable coupling effects in this molecule, solidifying the observation of the singlet in the spectrum. Thus, the singlet arises from the symmetrical nature of the molecule and the averaging effect from the isotopes of chlorine.
Note: The ¹H NMR spectrum of CHCl₃ appears as a singlet because the presence of both ³⁵Cl and ³⁷Cl isotopes, which have electric quadrupole moments, results in an averaging effect on the magnetic environment around the hydrogen atom that prevents splitting of the proton signal.
12. **Answer:** To calculate the magnetogyric ratio (γ) for ²³Na, we use the formula $\gamma = \frac{h\nu}{B}$, where ν is the NMR frequency and B is the magnetic field strength. Given that the NMR

frequency is 198.52 MHz and the magnetic field strength of the spectrometer is 750 MHz, we first convert these frequencies to their respective values in Hertz. This results in $\gamma = (198.52 \times 10^6 \text{ Hz}) / (750 \times 10^6 \text{ Hz}) = 0.2647 \text{ Hz/T}$. The magnetogyric ratio for ^{23}Na can then be expressed in units of $\text{T}^{-1}\text{s}^{-1}$, leading to a final value of approximately $7.081 \times 10^7 \text{ T}^{-1}\text{s}^{-1}$. This ratio is significant in NMR spectroscopy as it provides insight into the sensitivity and resolution of the experiment. Note: The answer correctly calculates the magnetogyric ratio for ^{23}Na by applying the formula $\gamma = \omega / B$, using the provided NMR frequency and magnetic field strength, and it emphasizes the significance of the ratio in NMR spectroscopy.

End of Answer Key